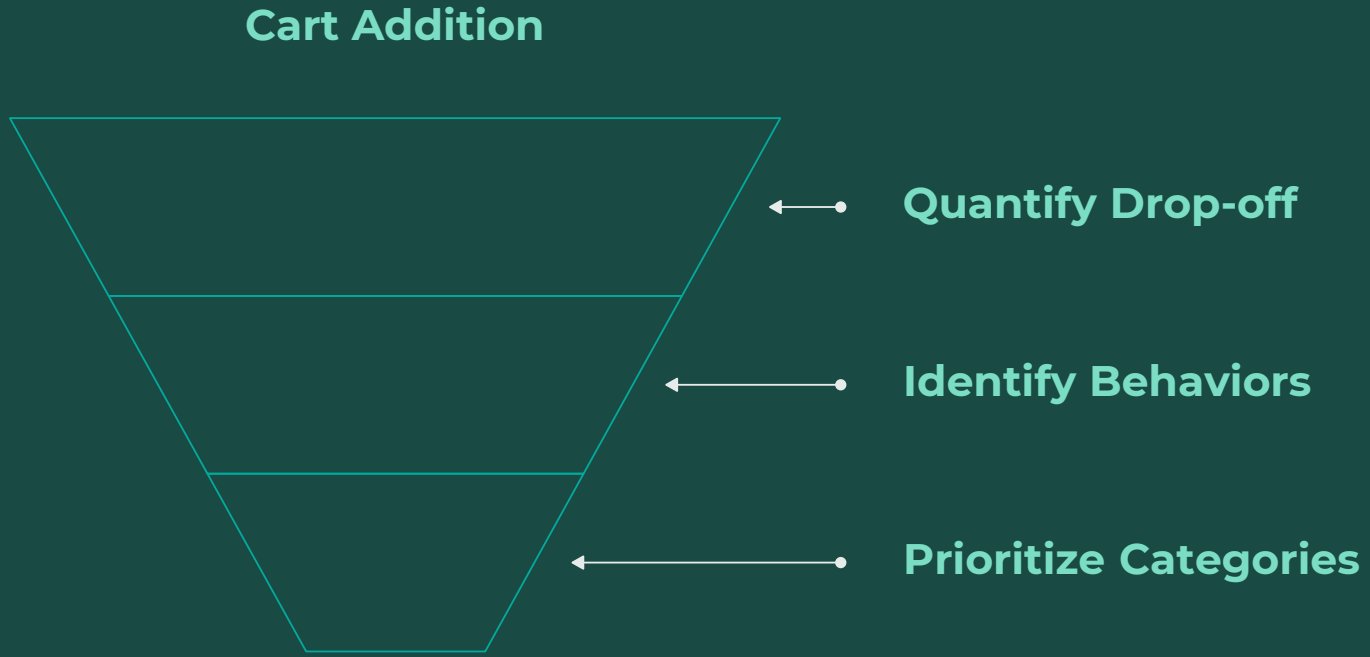
E-commerce Analysis

E-commerce Checkout Drop-off Analysis

SQL + Power BI | 9.24M Customer Sessions



E-commerce Checkout Drop-off Analysis



Analytics Workflow

A structured approach from raw data to actionable insights



01

Raw E-commerce Events

Ingesting session-level activity data.

02

PostgreSQL Validation & Indexing

Ensuring data integrity and query performance.

03

Session-Level Funnel Modeling

Building the conversion path logic.

04

Behavior & Category Analysis

Segmenting drop-offs by user actions and products.

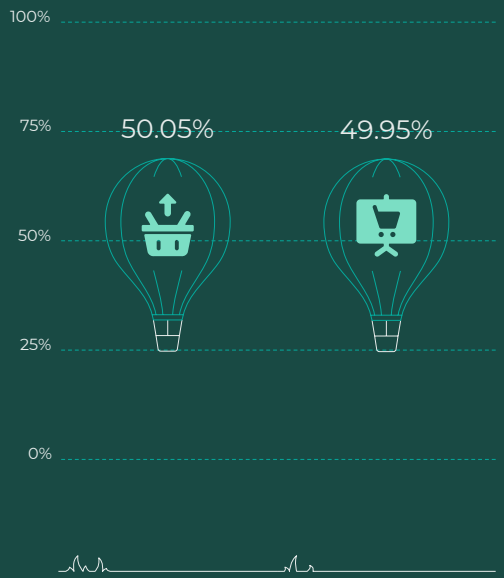
05

Power BI Dashboard & Business Actions

Visualizing results for executive decision-making.

Funnel Overview

Conversion Funnel



The data includes 8,344,836 sessions that were viewed only and 337,699 purchases made without a cart.

Behavioral Drivers

Key session characteristics associated with abandonment

53.72%

Single-item carts

Abandonment for single-item carts

61.29%

11+ product views

Abandonment for 11+ product views

59.11%

6+ repeat views

Abandonment for 6+ repeat views

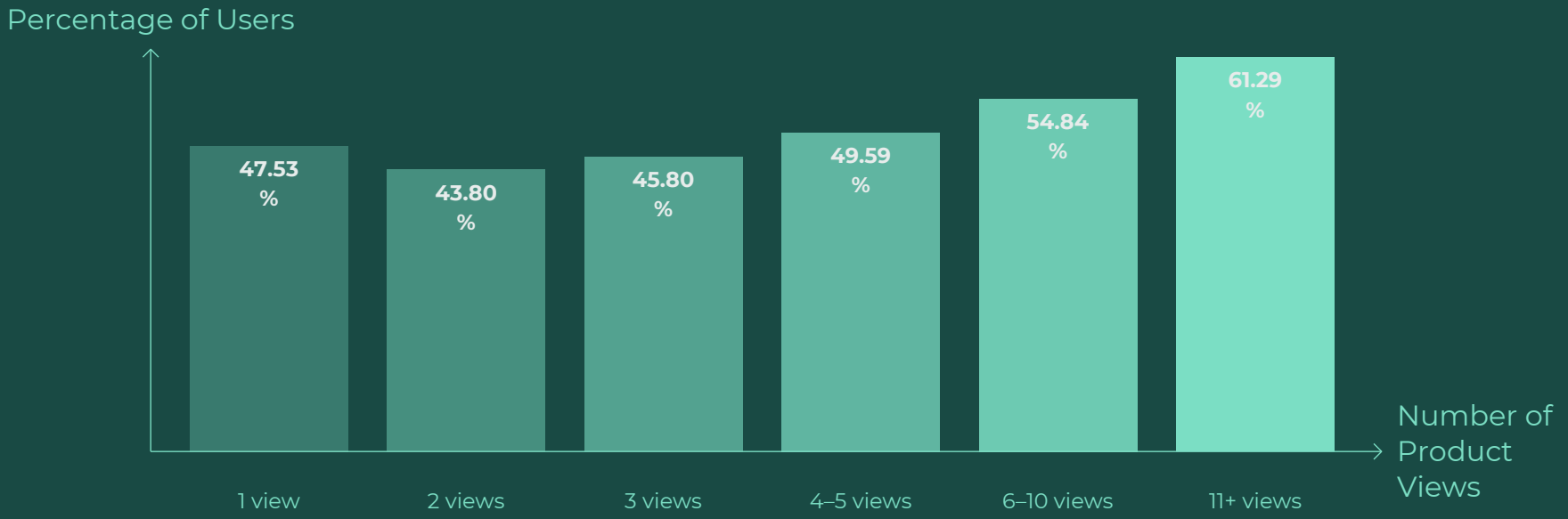
14.42 min

Average duration

Average duration for abandoned sessions (vs 10.53 min converted)

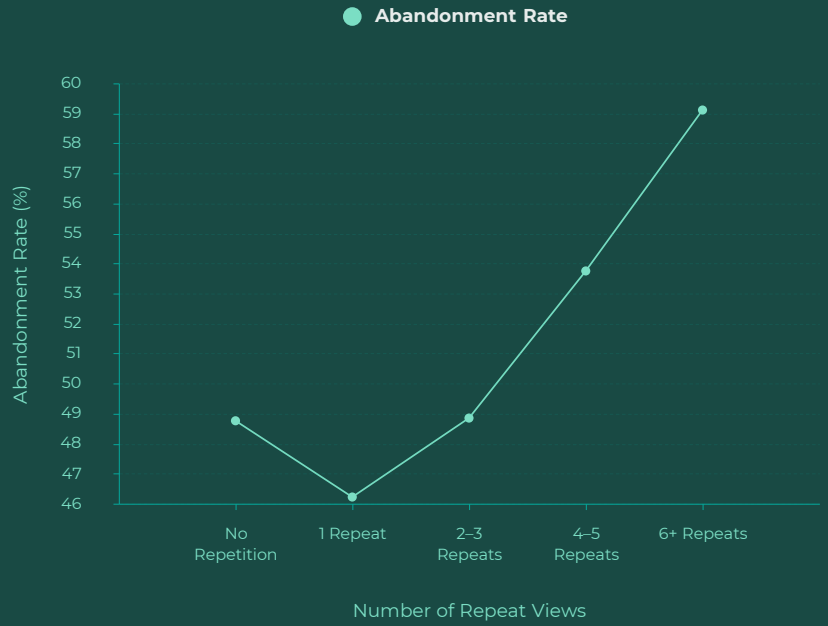
Product Exploration

High exploration may signal comparison difficulty or purchase hesitation

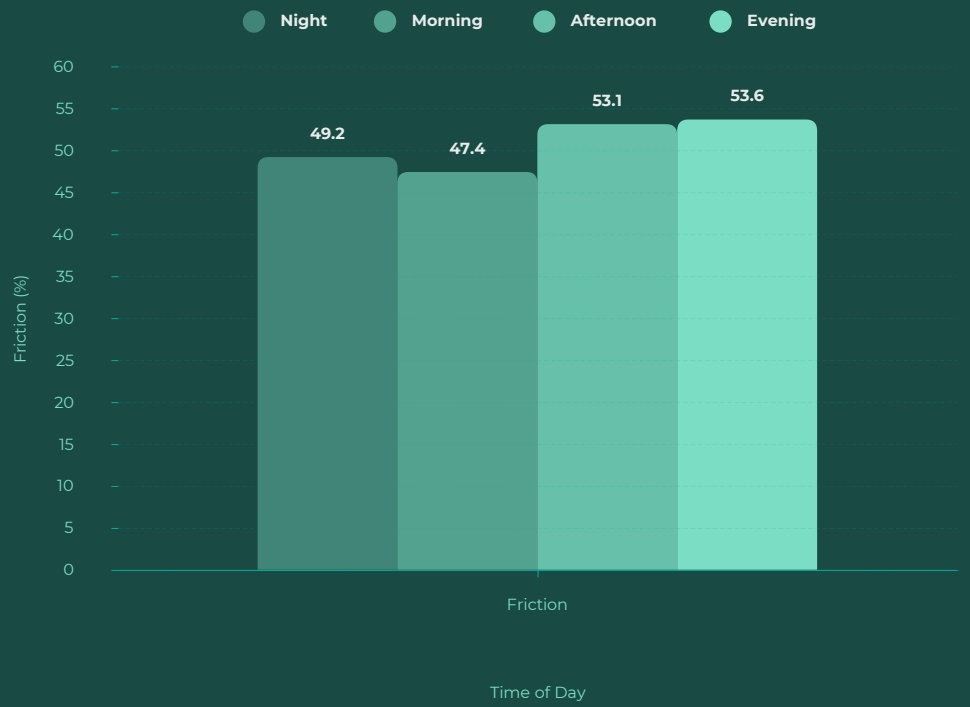


E-commerce Checkout Abandonment Rate by Repeat Views

The e-commerce checkout abandonment rate significantly increases for users with 4 or more repeat views, indicating a critical drop-off point for highly engaged customers.



E-commerce Checkout Drop-off by Time of Day





Category Opportunity Matrix

Prioritizing high-friction, high-volume categories for intervention

Action Area

High volume, high abandonment categories include computers.desktop, construction.tools.drill, appliances.kitchen.refrigerators, and appliances.kitchen.kettle.

Monitor

High volume, lower abandonment category includes electronics.smartphone at approximately 44.89%.

Roadmap to improve checkout completion



Simplify single-item checkout

Reduce steps for users with clear intent.

Add decision-support tools

Add comparison tools to reduce hesitation for high-exploration users.

Prioritize critical categories

Focus UX fixes on the 'Action Area' products.

Investigate evening friction

Audit site performance and traffic sources during peak hours.

Segment cart recovery

Tailor re-engage-ment emails based on session behavior.



Experiment Framework

Validating hypotheses through controlled A/B testing

Operating loop

Observation → Hypothesis → Intervention → A/B Test → KPI → Decision

Example

11+ views (61.29% abandonment) → Hypothesis: Comparison difficulty → Intervention: Test comparison feature → KPI: Cart conversion.

From Drop-off Data to Business Action

	Metric or Insight
Cart abandonment rate	50.05%
Primary indicators of user friction	High exploration & repetition
Category prioritization strategy	Balance volume, abandonment rate, and friction
Next step	Validate recommendations through controlled experiments